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DEEP-RX FOR HYPERSPECTRAL ANOMALY DETECTION

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ABSTRACT

Hyperspectral anomaly detection is a widely studied topic that has garnered significant attention in recent years. However, designing effective nonlinear detectors remains a challenge for many traditional methods. To address this issue, we propose the integration of a variational autoencoder (VAE) in this paper. The VAE enables efficient feature extraction from hyperspectral images (HSIs) by mapping inputs to latent variables that follow a Gaussian distribution. The resulting latent representations are subsequently passed to the Reed-Xiaoli (RX) detector to obtain the final detection results. Through extensive testing on three real datasets, the detection results demonstrate the superiority of our proposed method.

***Index Terms*—** Hyperspectral anomaly detection, variational autoencoder (VAE), RX detector

1. INTRODUCTION

Hyperspectral images (HSIs) are three-dimensional cubes that provide both spatial and spectral information, enabling the distinction between objects with different properties [1]. The abundance of spectral-spatial data in hyperspectral imaging has made it a prominent technology in various remote sensing tasks, including classification, detection, and anomaly detection. Hyperspectral anomaly detection (HAD) focuses on identifying pixels that deviate from the background spectra without any prior knowledge of target information [2, 3].

Among the different methods proposed for HAD, the RX detector, introduced by Reed and Yu [4], stands as a classical benchmark algorithm due to its simplicity, practicality, and robustness. The RX algorithm assumes a multivariate Gaussian distribution for the background and utilizes the Mahalanobis distance to measure the statistical difference between the pixel under consideration and the background. Building upon this approach, Kown et al. [5] propose the Kernel RX (KRX) anomaly detection method by incorporating kernel functions into the RX algorithm. While KRX provides a nonlinear mapping of the spectral signal, it faces challenges in kernel parameter selection, potentially violating the Gaussian assumption. Furthermore, KRX lacks dimensionality reduction and channel selection, resulting in increased processing time.

Recent advancements in deep learning have shown promising results in HAD tasks. Autoencoder (AE), an unsupervised learning method, reconstructs HSIs by training an encoder-decoder network [6, 7]. However, AE-based methods solely rely on the reconstruction error for anomaly detection, neglecting effective modeling of background statistics from the latent space. To address this limitation, the variational autoencoder (VAE) [8] has been proposed to estimate the potential probability distribution for each sample during the reconstruction process [9].

In this paper, we present a novel framework named DeepRX that combines VAE and the RX algorithm for hyperspectral anomaly detection. First, we leverage the VAE network to extract latent representations of input pixels, conforming to the characteristics of a Gaussian mixture model estimation. By employing the accurate and flexible VAE model instead of handcrafted kernel methods, we automatically extract comprehensive feature information. Subsequently, the classical RX algorithm is applied to detect abnormal values by analyzing the latent vectors, resulting in higher efficiency compared to direct pixel space analysis. Harnessing the non-linear expressive capabilities of neural networks, our proposed model extracts key features and adaptively generates Gaussian mixture probability distributions, ultimately enhancing the detection performance. Finally, we demonstrate the accuracy and efficiency of the DeepRX framework through extensive experiments conducted on multiple datasets.

2. PROBLEM FORMULATION

Suppose the hyperspectral image data cube is represented by $\mathbf{X} \in \mathbb{R}^{B\times N}$, where B represents the number of spectral channels and N represents the total number of pixels. $\boldsymbol{x}_i = [x_{1i}, x_{2i}, ..., x_{bi}]^T$ is the spectrum of each pixel.

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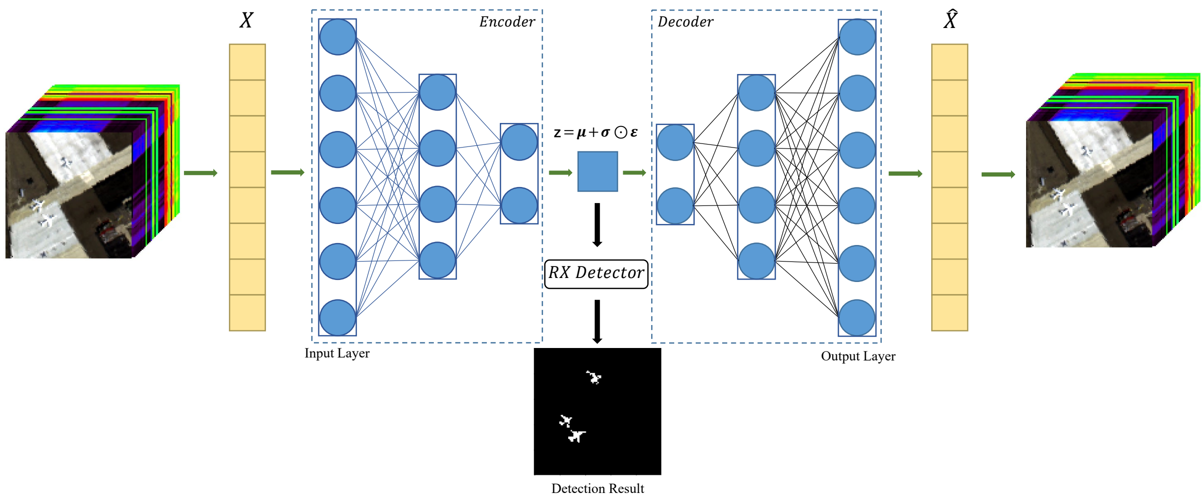

Fig. 1. The overall pipeline of the proposed DeepRX method.

In the RX anomaly detector, it is assumed that the background data information and the abnormal target information in the hyperspectral image follow Gaussian distributions with the same covariance but different means. The simplified form of the traditional RX detection is obtained using the generalized likelihood ratio test, given by:

$$\boldsymbol{r}=\left(\boldsymbol{r}-\hat{\boldsymbol{\mu}}_b\right)^{\mathrm{T}} \hat{\boldsymbol{C}}_b^{-1}\left(\boldsymbol{r}-\hat{\boldsymbol{\mu}}_b\right) \underset{H_0}{\overset{H_1}{\gtrless}} \boldsymbol{\eta}. \tag{1}$$

In the formula, $\boldsymbol{r}$ represents the spectral signal to be detected, $\boldsymbol{\eta}$ denotes the decision threshold, and $\hat{\boldsymbol{\mu}}_b$ and $\hat{\boldsymbol{C}}_b$ are the estimated values of the background mean and covariance matrix, respectively. These estimated values are given by:

$$\hat{\boldsymbol{\mu}}_b=\frac{1}{m} \sum_{i=1}^m \boldsymbol{x}_{\boldsymbol{i}} \tag{2}$$

$$\hat{\boldsymbol{C}}_b=\frac{1}{m} \sum_{i=1}^m\left(\boldsymbol{x}_i-\hat{\boldsymbol{\mu}}_b\right)\left(\boldsymbol{x}_i-\hat{\boldsymbol{\mu}}_b\right)^{\mathrm{T}} \tag{3}$$

By utilizing the nonlinear function Φ, each data point $\boldsymbol{x}_n$ from the original hyperspectral image space can be mapped to a feature space, denoted as $\Phi(\boldsymbol{x}_n)$. Accordingly, the proposed nonlinear RX algorithm can be expressed as follows:

$$\boldsymbol{r}=\left(\Phi(\boldsymbol{x})-\hat{\boldsymbol{\mu}}_{\Phi}\right)^T \hat{\boldsymbol{C}}_{\Phi}^{-1}\left(\Phi(\boldsymbol{x})-\hat{\boldsymbol{\mu}}_{\Phi}\right). \tag{4}$$

where $\hat{\boldsymbol{\mu}}_{\Phi}$ and $\hat{\boldsymbol{C}}_{\Phi}^{-1}$ are the estimate values of background mean and covariance matrix.

3. PROPOSED METHOD

The overall structure of the proposed DeepRX detector is illustrated in Figure 1. Our model primarily comprises two key components. Firstly, the VAE network is employed to obtain the Gaussian mixture probability distribution for each sample. Secondly, the extracted features are fed into the RX detector to obtain the final detection result.

3.1. VAE-based architecture

Similar to the structure of an autoencoder, the input data $\mathbf{X} \in \mathbb{R}^{B \times N}$ is first passed through the encoder network module. The encoder applies constraints to generate latent variables that follow a multivariate Gaussian distribution.

As shown, $\boldsymbol{z}$ is sampled from the encoder, conditioned on the input pixel data $\boldsymbol{x}$. This latent variable $\boldsymbol{z}$ contains the information of the input pixel data and satisfies the Gaussian distribution assumption required by the RX algorithm. Next, $\boldsymbol{z}$ is passed to the decoder structure. In the decoder network module, the final output aims to reconstruct the original data $\boldsymbol{x}$ as accurately as possible. In this network architecture, the loss function is defined as follows:

$$\mathcal{L}=\frac{\lambda_1}{N} \sum_{i=0}^N\left\|\boldsymbol{x}_i-\boldsymbol{x}_i^{\prime}\right\|_2^2+\lambda_2 D_{\mathrm{KL}}[q(\boldsymbol{z} \mid \boldsymbol{x}) \| p(\boldsymbol{z})] \tag{5}$$

where $\boldsymbol{x}_i^{\prime}$ represents the reconstructed pixel output generated by the decoder. DKL denotes the KL divergence, and λ_1 and λ_2 are hyperparameters. The first term of the loss function represents the reconstruction loss, which is computed using the mean squared error. The second term represents the KL divergence, measuring the difference between the distribution $q(\boldsymbol{z} \mid \boldsymbol{x})$ and the prior distribution $p(\boldsymbol{z})$. The prior distribution $p(\boldsymbol{z})$ follows a Gaussian distribution.

3.2. Deep RX detection framework

By utilizing the trained VAE network, we can obtain a feature mapping from $\boldsymbol{x}$ to $\boldsymbol{z}$. We input the data $\mathbf{T} \in \mathbb{R}^{B \times N}$ to be detected into our designed VAE network. Through the forward propagation in the encoder, we obtain the feature representation $\Phi(\boldsymbol{t}) \in \mathbb{R}^{D \times N}$, which consists of the latent variables for each sample data. By combining Equation (4), we can derive the abnormal results $\boldsymbol{r}$ corresponding to each pixel. The proposed framework is summarized in Algorithm 1.

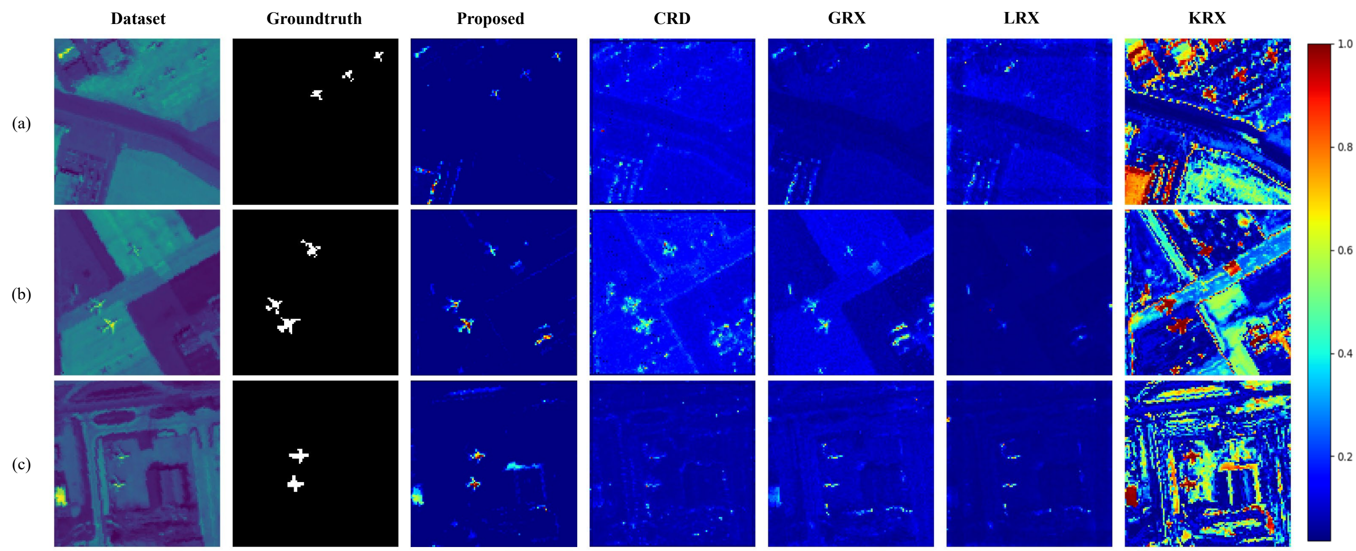

Fig. 2. Detection results of different methods (Proposed, CRD, GRX, LRX, KRX) on each dataset, (a) San Diego I, (b) San Diego II, (c) Los Angeles I

Algorithm 1 The proposed Deep RX framework

Input: Hyperspectral image $\mathbf{X}$
Output: Output of the detector.
1: Initialization. Set iteration number *epoch*, hyper-parameters λ_1, λ_2, random network weighets $\mathbf{W}$, latent space dimension *D*.
2: **For** $k = 1$: *epoch*
3: Forward propagation. Update the loss by eq. (5).
4: Back propagation. Update network weights $\mathbf{W}$.
5: **End For**
6: Save $\boldsymbol{z}$ using encoder.
7: Calculate anomaly result r using eq. (4).

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4. EXPERIMENTS

In this section, we conduct experiments on several hyperspectral datasets to evaluate the efficiency and advantages of our proposed method.

4.1. Experimental datasets

We utilize the following hyperspectral datasets for evaluation:
(1) San Diego I Dataset: This dataset, shown in Figure 2(a), was collected by AVIRIS at the San Diego airport area, CA, USA. It has a spectral resolution of 10 nm and a spatial resolution of 20 m. We remove bands with low signal-to-noise ratio (SNR) and water absorption, resulting in 189 remaining bands for our experiments.
(2) San Diego II Dataset: Shown in Figure 2(b), this dataset consists of a 100×100 image acquired from another dataset over the San Diego airport. The spectral wavelength ranges from 370 to 2510 nm with a spectral resolution of 10 nm. Similar to the previous dataset, we select information from 189 bands for evaluation. In both datasets, we consider airplanes as anomalies.
(3) Los Angeles I Dataset: Figure 2(c) displays the Los Angeles I dataset obtained by AVIRIS over Los Angeles, USA. The airport image covers an area of 100×100 pixels, and it contains 205 spectral bands with a spatial resolution of 7.1 m.

4.2. Experimental results

We compare our method with four algorithms that performed well in previous studies: Global RX [4] (GRX), Local RX [10] (LRX), Kernel RX [5] (KRX), and CRD [11]. For LRX and CRD, we conducted a series of experiments and found that setting the inner window size to 10 and the outer window size to 20 yielded optimal results. We used a Gaussian kernel with a width of $s = 0.1$ to achieve the best performance for KRX.

In our proposed method, we set the latent space dimension to 10 and used three hidden layers with dimensions [128, 64, 32]. The hyperparameters λ_1 and λ_2 were respectively set to 0.1 and 0.9. The network was trained for 300 epochs on a GeForce GTX 2080Ti GPU.

The detection results of all methods on different datasets are shown in Figure 2. Comparing with the ground truth, our proposed method exhibits the smallest number of false alarms while effectively detecting anomalies. In contrast, KRX shows relatively higher false alarm rates. Furthermore, the detection results demonstrate that CRD, GRX, and LRX are ineffective in effectively separating anomalies and background pixels across all datasets.

Receiver Operating Characteristic (ROC) curves provide a quantitative analysis of the relationship between detection

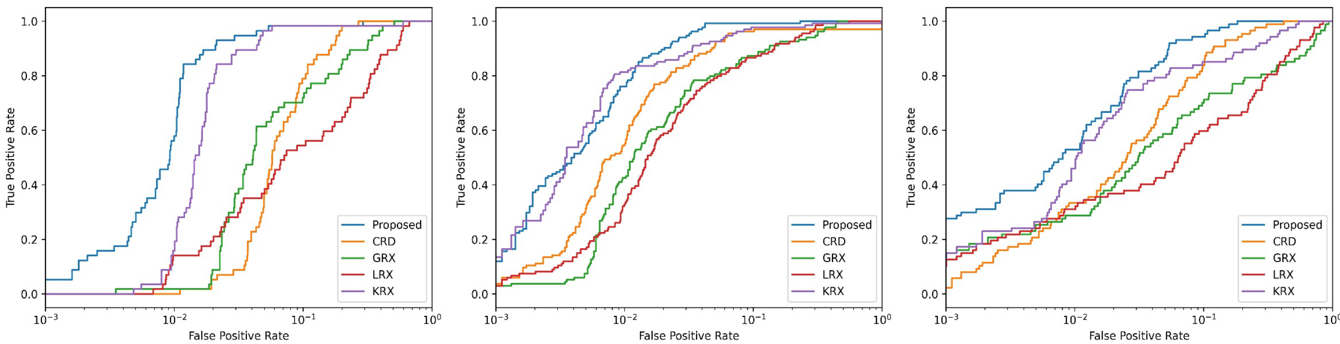

Fig. 3. ROC curves of different methods for the three real datasets. (a) San Diego I, (b) San Diego II, (c) Los Angeles I.

Table 1. AUC values of different methods on three datasets

Dataset	Proposed	CRD	GRX	LRX	KRX
San Diego I	**0.991**	0.956	0.952	0.953	0.986
San Diego II	**0.985**	0.923	0.905	0.825	0.972
Hyper Urban	**0.979**	0.949	0.840	0.832	0.940

probability and false alarm probability, with the Area Under the Curve (AUC) serving as an important criterion for assessing detection performance. The closer the AUC value is to 1, the better the algorithm's performance. The ROC curves of all methods are depicted in Figure 3, where the ROC curve of our proposed method generally outperforms the other methods. Additionally, the corresponding AUC values for the detection results are provided in Table 1. Our method achieves the highest scores on all three datasets, with values of 0.991, 0.985, and 0.979, respectively. In terms of computational performance, our algorithm has a slower detection speed compared to the vanilla RX algorithm, but it is up to ten times faster than LRX and KRX.

5. CONCLUSION

In this paper, we presented the DeepRX method for hyperspectral anomaly detection. By leveraging a neural network, we obtained feature representations that fully exploited the nonlinear information contained in the hyperspectral image bands. These features were then fed into the RX detector to obtain the final detection result. Experimental results demonstrated that the proposed DeepRX outperformed several other state-of-the-art anomaly detection methods.

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